

Syllabus

Course Information

Course Number: ECEN 629
Course Title: Applied Convex Optimization
Section: 600, 700
Time: T Th 12:45-2:00 PM
Location: ETB 1003
Credit Hours: 3

Instructor Details

Instructor: Dr. Chao Tian
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Office Hours: TBA

Course Description

Introduction to convex optimization including convex set, convex functions, convex optimization problems, KKT conditions and duality, unconstrained optimization, and interior-point methods for constrained optimization; applications in information science, digital systems, networks and learning.

Course Prerequisites

Linear algebra, vector space methods, calculus, probability (ECEN 601 or equivalent)

Course Learning Outcomes

At the end of the course students will be able to:

- *Understand mathematical optimization problems*
- *Know the difference between different classes of optimization problems*
- *Formulate mathematical optimization problems*
- *Have basic ability to convert problems into simpler forms*
- *Understand and apply duality properties for convex problems*
- *Understand the necessity and sufficiency of KKT conditions*
- *Understand basic concepts of convergence behavior of the numerical optimization algorithms*
- *Understand the algorithms to solve optimization problem and choose suitable ones*

Textbook and/or Resource Materials

Required: Convex optimization, Boyd and Vandenberghe, Cambridge University Press, 2004 (available free online at authors' webpage)

Grading Policy

- 5 homework sets: 10%
- Quizzes: 5%
- First Midterm (3/10): 25%
- Second midterm (4/21): 25%
- Final exam (cumulative with focus on the second half): 30%
- Must complete your own homework (discussion with classmates is allowed)
- Grading scale: A = 90-100, B = 80-89, C = 70-79, D = 60-69, F = <60

Class Participation – Active participants 5%

AI Usage Policy – It is not allowed to use LLMs' answers as your homework solutions. You can, however, try to consult LLMs for specific concepts or for verification of your own answers. You must indicate clearly on your hand-ins that LLMs/AI were used, though usage according to the rule above will not be penalized. Warning: Historically, students who use LLMs extensively for their homework tend to do poorly in the exams.

Electronic Device Policy – During class, cellphones and laptops must be stored, unless otherwise instructed. You can use either a stylus (e-pen) to take notes on Ipad (or Kindle, or equivalent android devices). A real pen-notebook pair is highly encouraged.

Late Work Policy

- Late homework will be accepted subject to 20% penalty for each 24 hours;
- Late exam paper will not be accepted.

Work submitted by a student as makeup work for an excused absence is not considered late work and is exempt from the late work policy. (See Student Rule 7.)

Course Schedule

Week	Topic	Required Reading
1	Introduction to mathematical optimization, linear vs. non-linear optimization	pp. 1-11
2-3	Convex set, operations that preserve convexity, separating and supporting hyperplanes, cones, and dual cones (1 st HW)	pp. 19-51
4-5	Convex functions, properties, operations that preserve convexity, the conjugate function, quasi-convex functions, log-convex functions	pp. 67-107
6-7	Convex optimization: basic form, linear program, quadratic optimization, geometric programming (2 nd HW)	pp. 127-167
8-9	Duality: Lagrange dual functions, dual problem, optimality conditions, sensitivity analysis (1 st midterm)	pp. 215-258
10	Approximation and fitting: norm approximation, least-norm, regularized approximation (3 rd HW)	pp. 291-324
11	Statistical estimation: parameter estimation, supporting vector machine	pp. 351-384
12	Resource allocation in communication systems	Supplementary material

13	Algorithms for unconstrained minimization (2 nd midterm)	pp. 457-496
14	Equality-constrained minimization (4 th HW)	pp. 521-542
15	Introduction to interior-point methods	pp. 561-579

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Notice of Nondiscrimination

Texas A&M University is committed to providing safe and non-discriminatory learning, living, and work environments for all members of the University community. The University provides equal opportunity to all employees, students, applicants for employment or admission, and the public regardless of race, color, sex (including pregnancy and related conditions), religion, national origin, age, disability, genetic information, or veteran status. Texas A&M University will promptly, thoroughly, and fairly investigate and resolve all complaints of discrimination, harassment (including sexual harassment), complicity and related retaliation based on a protected class in accordance with System Regulation 08.01.01, University Rule 08.01.01.M1, Standard Administrative Procedure (SAP) 08.01.01.M1.01, and applicable federal and state laws. In accordance with Title IX and its implementing regulations, Texas A&M does not discriminate on the basis of sex in any educational program or activity, including admissions and employment. The following person has been designated to handle inquiries and complaints regarding the non-discrimination policies: Jennifer M. Smith, TAMU Associate VP & Title IX Coordinator at YMCA Ste 108, College Station, TX 77843, 979-458-8407, or email civilrights@tamu.edu. For other reporting options, visit <https://ocrcas.ed.gov/contact-ocr> to locate the address and phone number of the office that serves your area, or call 1-800-421-3481.

Civil Rights, Free Speech, and Title IX Policies

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit discrimination and harassment based on an individual's race, color, sex, (including pregnancy and related conditions), religion, national origin, age, disability, genetic information, veteran status, or any other legally protected characteristic. This includes forms of sex-based violence, such as sexual assault, sexual harassment, sexual exploitation, dating/domestic violence, and stalking.

Students can report discrimination/harassment, access supportive resources, or learn more about their options for resolving complaints on the [University's Civil Rights & Title IX webpage](#).

Students should be aware that all university employees (except medical or mental health providers) are mandatory reporters, which means that if they observe, experience or become aware of an incident that they reasonably believe to be discrimination/harassment alleged to have been committed by or against a person who was a student or employee at the time of the incident, the employee must report the incident to the university.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below). Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

To request academic accommodations, contact the designated ADA office based on your location:

- Texas A&M University, College of Nursing, College of Dentistry, Irma Lerma Rangel College of Pharmacy College Station, College of Medicine, School of Public Health, Institute of Biosciences and Technology, EnMed Program, Bush School in Washington DC, Mays Business School – CityCentre, TAMU Engineering Academies, Texas A&M University Higher Education Center at McAllen and Texas A&M University at Galveston should contact [Disability Resources](#) at (979) 845-1637 or disability@tamu.edu.
- Texas A&M University School of Law should contact the Office of Student Affairs at (817) 212-4111 or law-disability@law.tamu.edu to request accommodations.
- Irma Lerma Rangel College of Pharmacy in Kingsville should contact the Disability Resource Center at Texas A&M University - Kingsville at (361) 593-3024 or drc.center@tamuk.edu to request accommodations.
- Texas A&M University College of Veterinary Medicine & Biomedical Sciences in Canyon should contact the Office of Student Accessibility at West Texas A&M University – Canyon at (806) 651-2335 or osa@wtamu.edu.
- Texas A&M University at Qatar (TAMUQ) should contact the campus psychologist, Dr. Steve Wilson +974-4423-0047 or stephen.wilson@qatar.tamu.edu.

If you are experiencing difficulties with your approved accommodations, contact the office responsible for approving your accommodations or the Texas A&M ADA Coordinator Julie Kuder at ADA.Coordinator@tamu.edu or (979) 458-8407.

Pregnancy Accommodations

Texas A&M provides reasonable accommodations to students due to pregnancy and/or related conditions, such as childbirth, recovery and lactation. Students should contact the University's [Pregnancy Coordinator](#) as soon as they become aware of the need for accommodation. Depending on the circumstances, accommodations could include extended time to complete assignments or exams, changes in course sequence, or modifications to the physical classroom environment. Texas A&M will also allow a voluntary leave of absence, ensure the availability of lactation space, and maintain grievance procedures to provide for the prompt and equitable resolution of complaints of sex discrimination. For information regarding pregnancy accommodations, email TIX.Pregnancy@tamu.edu.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through [University Health Services](#) on its [mental health webpage](#). The [TELUS Health Student Support app](#) provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org.

Students needing a listening ear can contact University Health Services (979.458.4584) 24-hour emergency help is also available through the 988 Suicide & Crisis Lifeline (988) or at 988lifeline.org.

Texas A&M at Galveston

Students who need someone to talk to can call (409) 740-4736 from 8:00 a.m. to 5:00 p.m. weekdays or visit tamug.edu/counsel for more information. For 24-hour emergency assistance during nights and weekends, contact the TAMUG Police Dept at (409) 740-4545. 24-hour emergency help is also available through the 988 Suicide & Crisis Lifeline (988) or at 988lifeline.org.

Statement on the Family Educational Rights and Privacy Act (FERPA)

FERPA is a federal law designed to protect the privacy of educational records by limiting access to these records, to establish the right of students to inspect and review their educational records and to provide guidelines for the correction of inaccurate and misleading data through informal and formal hearings. Currently enrolled students wishing to withhold any or all directory information items may do so by going to howdy.tamu.edu and clicking on the "Directory Hold Information" link in the Student Records channel on the MyRecord tab. The complete [FERPA Notice to Students](#) and the student records policy is available on the Office of the Registrar webpage.

Items that can never be identified as public information are a student's social security number, citizenship, gender, grades, GPR or class schedule. All efforts will be made in this class to protect your privacy and to ensure confidential treatment of information associated with or generated by your participation in the class.

Directory items include name, UIN, local address, permanent address, email address, local telephone number, permanent telephone number, dates of attendance, program of study (college, major, campus), classification, previous institutions attended, degrees honors and awards received, participation in officially recognized activities and sports, medical residence location and medical residence specialization.

Free Speech and Civil Discourse

Texas A&M recognizes that the pursuit of truth through open and robust discourse is critical to academic inquiry. However, as a community of scholars, the university has an aspirational expectation that such discourse will be conducted in accordance with Aggie Core Values. In this "marketplace of ideas," we encourage civil dialogue creating an environment that allows individuals to express their ideas and to have their ideas challenged in respectful and responsible ways. Students can learn more about Freedom of Expression and Free Speech on the University's [website](#) about the [First Amendment](#).

Artificial Intelligence Statements

With the emergence of artificial intelligence (AI) technologies, the ways in which we define our creative processes continue to transform. AI generators are rapidly evolving from simple editing for grammatical errors and spelling mistakes (Grammarly, MS Word Spell Check) to sophisticated text production (ChatGPT, Google Bard, etc.), as well as image, computer code, and audio generation. The presence of such tools, however, does not replace our need to learn how to draft, revise, and reflect on texts, programs, drawings and how to exercise information literacy and personal responsibility in how we locate, evaluate, incorporate, and cite primary/ secondary sources. For example, the Association for Writing Across the Curriculum states the following: Writing to learn is an intellectual activity that is crucial to the cognitive and social development of learners and writers. This vital activity cannot be replaced by AI language generators (AWAC).